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Discovering the Wormhole: Gravitational Waves

How might humans have discovered *Interstellar*'s wormhole? As a physicist, I have a favorite way. I describe it here in an extrapolation of *Interstellar*'s story—an extrapolation that, of course, is my own and not Christopher Nolan's.

LIGO Detects a Burst of Gravitational Waves

I imagine that decades before the movie begins, when Professor Brand was in his twenties, he was deputy director of a project called LIGO: The Laser Interferometer Gravitational Wave Observatory (Figure 16.1). LIGO was searching for ripples in the shape of space arriving at Earth from the distant universe. These ripples, called gravitational waves, are produced when black holes collide with each other, when a